

Liver-Chip Evidence Auditor: Albumin Response Review and Follow-up Planning

Submission category: Tool & Platform

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Project summary

Liver-Chip Evidence Auditor is a reproducible prototype that combines dose-response modelling with a compact local language model to help researchers review organ-on-chip experiments and choose a follow-up. We propose a workflow laboratories can adapt to their own assays and operating constraints, demonstrated here with published liver-chip albumin data.

Researchers review the response, choose a feasible action, and record what happened. The statistical auditor fits the dose-response curve, tests whether individual readings or dose conditions change the conclusion, and identifies the relevant source records. The optional 135M language model selects an action using the audit, researcher request and available resources. Code supplies the numerical evidence and source references, checks action eligibility, and provides a rules fallback.

The demonstration includes six compounds, 35 dose conditions and 68 observed readings. For Clozapine, temporarily omitting one reading changes the result from no in-range 50% crossing to a crossing near 256.22 times unbound Cmax. The app identifies ALBUMIN! C12 for inspection while retaining the original reading. Researchers can then record their findings, compare a documented compatible data revision with the baseline, and export the review.

Laboratories can reproduce the example using the supplied code, source data and optional model checkpoint. They can use the implementation as a starting point for their own data mappings, endpoint models, thresholds and follow-up policies. Researchers can retain the same sequence from source measurements to a documented follow-up while adapting the scientific assumptions to their assay.

Materials and submission status

[Public code repository](#).

Public release: [code package](#), [optional checkpoint](#) and [technical report](#).

The [app walkthrough](#) follows the published Clozapine example through source inspection, resource selection, outcome recording, export and a local-model request. The Kaggle card is supplied as [submission/kaggle_card_560x280.png](#).

Prototype workflow and reference assay

This submission provides a reproducible reference implementation of three connected tasks: review a dose-response conclusion, select a follow-up under laboratory resource constraints, and record the outcome against the original evidence. Laboratories can use the code to develop their own data mappings, endpoint models and action policies while preserving this sequence.

The implemented reference assay is day-three liver-chip albumin. A scientist reviews whether a 50% normalised response crossing is supported inside the measured dose range and identifies the readings or assumptions that affect that conclusion. Compatible albumin CSVs can be imported through the local app. The laboratory-adaptation section specifies the code changes and validation needed for other assays.

Albumin production provides a functional readout in the source Liver-Chip study. This workflow focuses on the released day-three albumin responses and their operational 50% crossing. The source study combines albumin with other endpoints for its broader toxicology assessment. [Ewart et al., 2022](#).

The three published drug comparison pairs provide practical examples. The response workbook identifies compound, dose and endpoint records; chip, donor and cycle identities are undocumented, so the analysis compares compound responses without pairing independent chips.

Research context and contribution

Reliable organ-on-chip interpretation requires a defined context of use, adequate controls and experimental provenance. Fit-for-purpose MPS quality recommendations motivate this emphasis on what the available evidence permits us to conclude. [Pamies et al., 2024](#).

The drc framework supports model specification and effective-dose estimation; the original Liver-Chip paper used its log-logistic modelling approach. Control-aware Hill fitting addresses half-maximal concentration estimates for incomplete curves. These methods establish the modelling context for our source-linked review workflow. A laboratory adaptation with raw reference measurements could incorporate control-aware inference. [Ritz et al., 2015](#), [La et al., 2023](#).

An automated platform for developmental neurotoxicity data examines statistical decisions and flags data outside its applicability domain. It provides conceptual prior work for automated evidence review. Here, the implementation and evaluation focus on liver-chip albumin responses and their source records. [Keßel et al., 2023](#).

The contribution is the implemented connection between statistical evidence review, constrained language-model action selection and researcher follow-through. The statistical code calculates the response and sensitivity; the model selects an action class from the audit, request and resources; eligibility checks and an evidence renderer supply the delivered card. The researcher records findings or submits a documented evidence revision. Source records and retained evidence versions let another reviewer inspect the decision. The evaluation measures crossing sensitivity, warning performance and action-policy agreement separately.

Data and provenance

The core files are Supplementary Data 8 and the dosing context in Supplementary Data 1 of Ewart et al. The article states CC BY 4.0; attribution and a link to the licence accompany the data. Original workbooks are preserved, and every extracted reading retains its DOI, filename, worksheet, value cell and source-record ID. [Source paper, CC BY 4.0](#).

Quantity	Available evidence
Compound curves	6
Drug-dose conditions	35
Albumin source readings	70
Observed albumin readings	68
Missing albumin readings	2, both in pioglitazone
Dose scale	Multiple of unbound Cmax
Response scale	Source-normalised albumin percentage
Chip, donor, run and cycle mapping	Unavailable in this response subset
Raw vehicle controls	Unavailable in this response subset

The complete response extraction contains 210 scalar readings across albumin, ALT and morphology, with six missing values. Only albumin enters the model. These counts must not be described as independent chip counts. ALT's unresolved source units and the absence of a pairing key prevent joint endpoint modelling. Source-record IDs name cells; they cannot create experimental independence.

Synthetic records carry `data_origin=synthetic`; their chip and run IDs identify simulator entities. Published records retain unknown experimental identities, and real-data validation excludes synthetic values.

Implemented method

Source checks

Before fitting, the tool checks positive finite doses, finite observed responses, day-three albumin and its units, unique non-empty source-record IDs and whether an input is an original measurement. Mixed-compound curves and incompatible inputs block the analysis. Missing readings generate review information; missing chip/run mapping limits group validation. A cross-endpoint join requires a documented shared sample ID and is rejected when that evidence is absent.

The implementation treats equal numerical readings as distinct observations when their source cells differ. It does not silently fix source data or count derived summaries as additional measurements.

Statistical models

The primary model uses isotonic regression to fit a non-increasing response to dose means, weighted by observed counts. Let J be the number of dose conditions with at least one observed reading, indexed in ascending dose order. At dose d_j , the observed mean is m_j , the observed count is n_j , and the fitted response is z_j . The estimate solves

$$\hat{\mathbf{z}} = \arg \min_{\mathbf{z} \in \mathbb{R}^J} \sum_{j=1}^J n_j (z_j - m_j)^2$$

subject to

$$z_j \geq z_{j+1}, \quad j = 1, \dots, J-1.$$

The pool-adjacent-violators algorithm merges adjacent increasing blocks using their observed counts. Flat fitted blocks are allowed. Between tested doses, prediction uses linear interpolation in $\log_{10}$ dose. The weights count observed readings; they do not establish independent chips or calibrated uncertainty. The fitted response levels are distinct from the adapted language-model parameters.

The conventional comparator is a four-parameter decreasing logistic curve. It fits lower response, amplitude, log-dose midpoint and slope by bounded nonlinear least squares with three starting points. The bounds and fit diagnostics are documented in `evaluation_protocol.md`. Failed fits, active bounds and poor Jacobian conditioning remain visible. The comparator's midpoint is not automatically the crossing of an absolute 50% normalised-response threshold.

Raw mean interpolation and a constant training mean supply simple baselines. Parameters are fitted from the available endpoint values; review rules and scenario choices are fixed. The optional action model is evaluated separately.

Crossing and sensitivity

The primary conclusion has three states. A finite downward crossing is estimated within the tested range. A response already at or below 50% at the lowest tested dose is left-censored under the boundary convention. Otherwise, if no downward crossing is reached within the tested range, the result is right-censored. Censored results have no finite crossing estimate; right censoring does not assert that an eventual crossing exists. Raw interpolation reports the first downward crossing and flags repeated downward crossings or a return above 50%. The 50% threshold is an operational endpoint reference, not a validated clinical margin.

For each curve, the tool compares alternative models, removes each reading where another observed reading remains at that dose, removes each dose condition when at least two doses remain, and probes missing readings with hypothetical values of 0, 50, 100, 150 and 200%. Singleton-dose influence is covered by eligible dose-deletion scenarios. The probes are neither imputations nor biological bounds. A material change is a different crossing/censoring state or a greater-than-twofold crossing-location shift. Individual scenarios and continuous location shifts are exported.

The revised combined review rule flags any material scenario change or missing response. Failed logistic comparisons count as material model scenarios. Standalone parameter bounds and weak identification remain numerical diagnostics and no longer trigger sensitivity review alone. The original diagnostic-inclusive rule is preserved as a comparator. Agreement across this finite set is not a probability of correctness.

Evidence checklist and observation influence

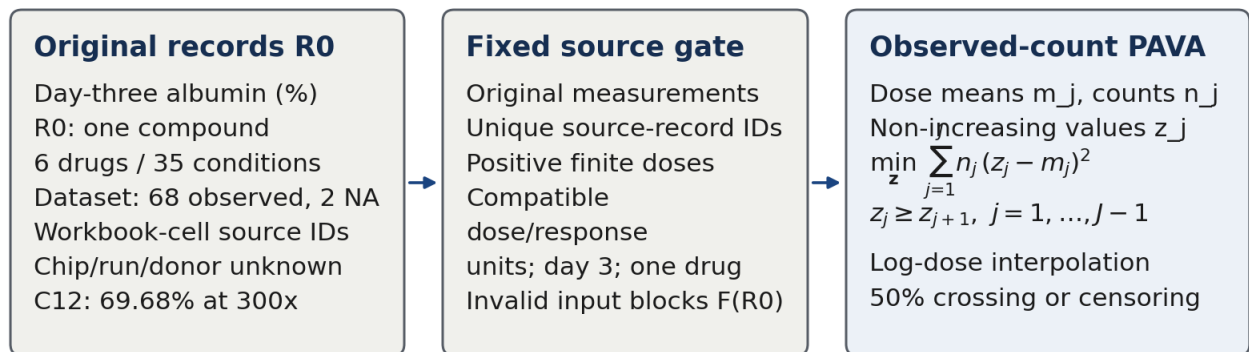
The checklist separates input validity, available experimental evidence, conclusion sensitivity and numerical diagnostics. Each item reports Pass, Fail or Not assessed, its supporting evidence and a next action. A pass concerns that named check only. Raw vehicle/reference controls, assay calibration, device QC and biological independence remain Not assessed because this release cannot verify them. Reported identity fields do not prove independence. A failed threshold-bracketing check requires a censored result rather than an unsupported finite estimate.

Observation-deletion scenarios retain source-record IDs, workbook cells, original values and both endpoint conclusions. The interface highlights readings whose temporary deletion changes the operational conclusion. Original observations remain in fitting and exports. These results measure influence; they do not establish erroneous outliers. Technical exclusion would require separate documented evidence. With only two readings per dose and undocumented independence, a residual-based label cannot resolve which reading is erroneous. This checklist adapts the fit-for-purpose QC approach described by [Pamies et al., 2024](#); it is not a substitute for laboratory qualification.

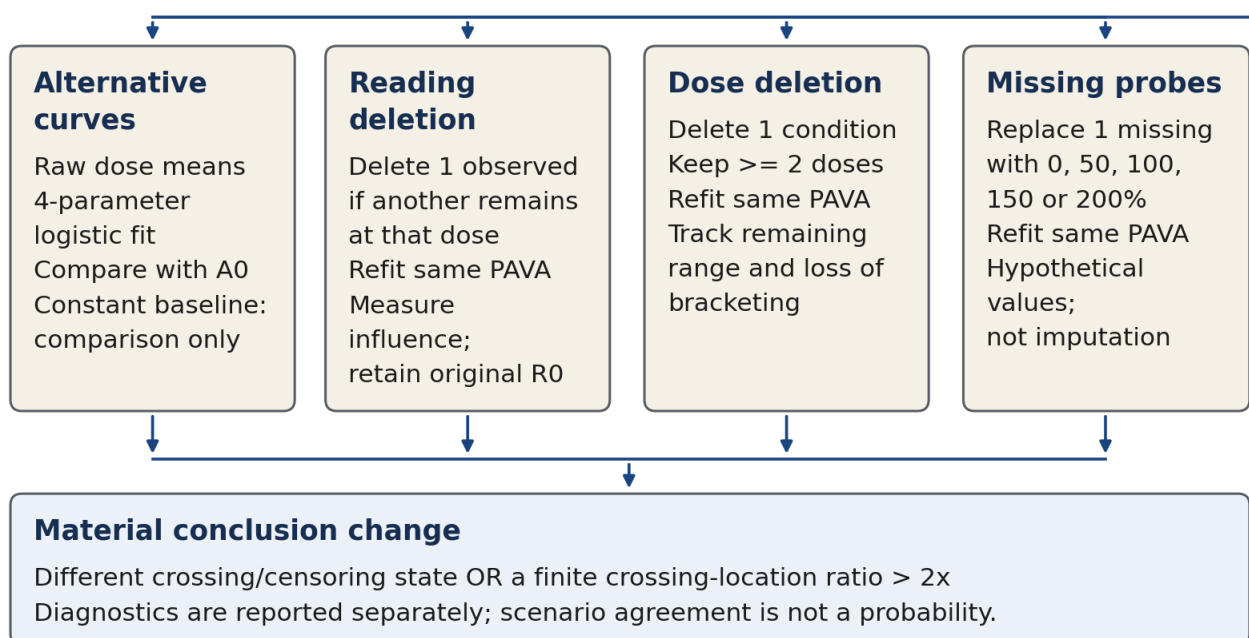
Liver-Chip Evidence Auditor

Figure 1. Original endpoint evidence → numerical inference → source-linked concerns

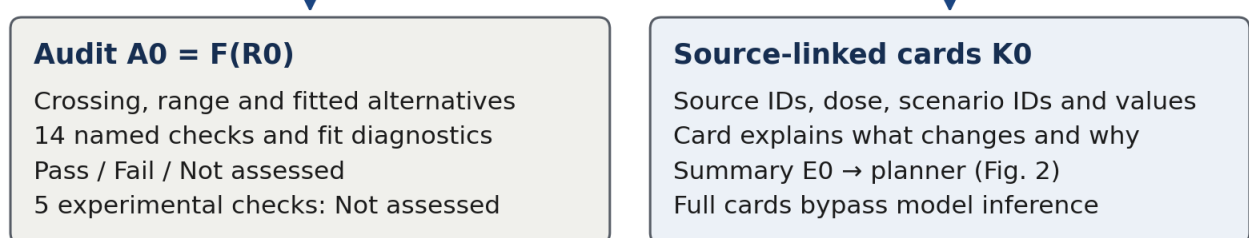
A. Per-compound source gate and fitted response



B. Finite sensitivity around the original fit A0



C. Audited evidence and deterministic source cards



Real Clozapine example: full fit is right-censored; temporary deletion of C12 gives 256.22x unbound Cmax. Influence does not establish an erroneous reading.

Figure 1. Source gate, weighted non-increasing fit and finite sensitivity. Each eligible scenario is compared with the original fit; source-linked cards retain the responsible records. The Clozapine deletion example uses published evidence and leaves the original reading in the baseline.

Framework objects and evidence boundaries

The implemented framework uses five explicit objects. R_0 is the original per-compound record set; $A_0 = F(R_0)$ is the audit computed by the numerical functions; K_0 contains the deterministic source/scenario-linked cards; E_0 is the reduced summary passed to action selection; and P_0 preserves the action together with its original evidence. Figures 1-3 show these objects across numerical inference, action-code selection and documented follow-through. The editable sources and a function-by-function verification map accompany the report in `framework_notes.md`.

Each sensitivity scenario reruns the relevant fit without modifying R_0 . Raw interpolation and logistic fitting enter model-sensitivity comparisons; the constant baseline is used only for prediction evaluation.

Action selection consumes E_0 , a researcher request q and resource choices r . Full source cards bypass language-model inference. The renderer selects an existing K_0 card after the delivered action is known.

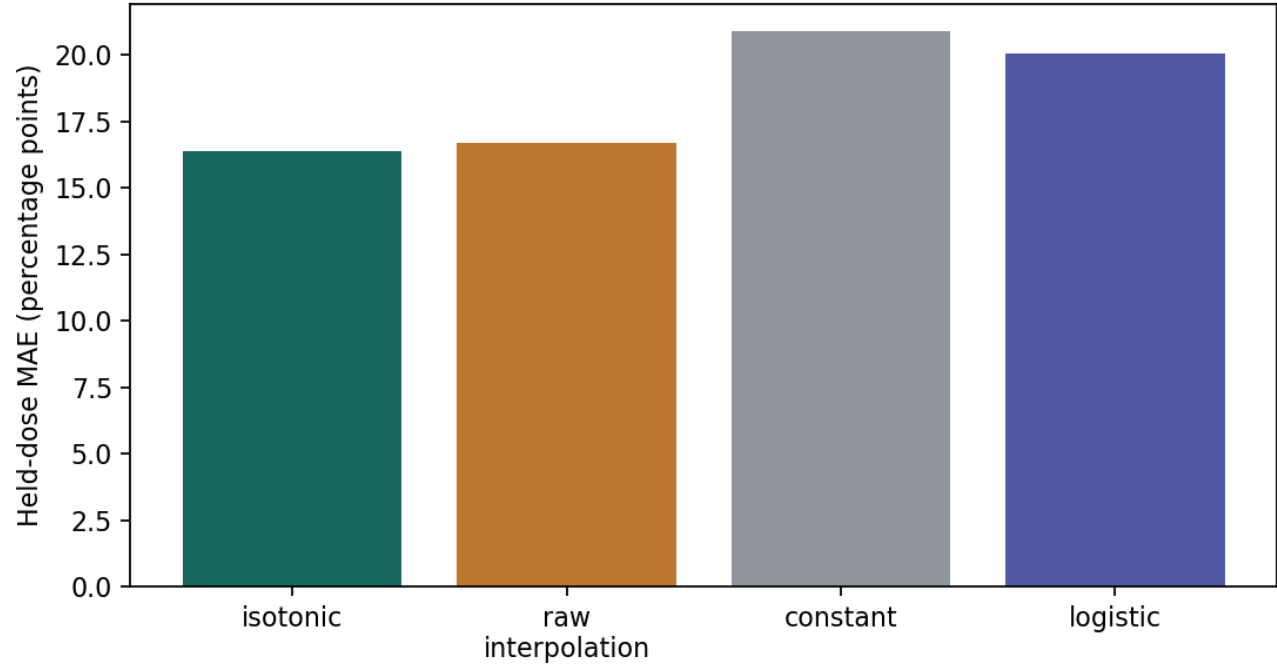
Real-data evaluation

The original protocol was frozen on 2 October 2026. The warning revision was specified on 4 October, after inspecting the earlier results and before implementing the revised rule. All comparisons on the real data and earlier simulation seeds are retrospective. At each of 23 interior dose conditions, all readings at that dose are withheld together. Other doses of the same compound remain in training. The held observed mean is used only for scoring; no held-dose reading enters fitting or warning computation. Boundary doses are excluded from this interpolation benchmark but included as deletion-sensitivity cases.

Model	MAE, percentage points	RMSE, percentage points	Eligible folds
Weighted isotonic	16.36	22.16	23/23
Raw mean interpolation	16.69	22.83	23/23
Constant training mean	20.89	24.29	23/23
Four-parameter logistic	20.07	28.15	23/23

Isotonic regression has the lowest MAE and RMSE in this reference dataset, with a 0.33-point MAE difference from raw interpolation. This comparison supports its use as the prototype's primary fit for these sparse curves. The 23 folds share compounds, so the result describes within-compound interpolation rather than statistical significance or performance on new drugs.

Same 23 interior dose conditions; correlated within six drugs



Held-dose comparison on identical conditions

To evaluate warnings, we compare the training-only conclusion with its counterpart after restoring the withheld condition. A change in state or a greater-than-twofold location shift is the reference event. The restored conclusion is still based on sparse data; it is not biological ground truth.

Warning method	Detected changes	Missed changes	Warnings without a reference change	Cases with no warning
No warning	0	1	0	23/23
Missingness / count warning	0	1	4	19/23
Source-only warning	0	1	4	19/23
Original diagnostic-inclusive combined warning	1	0	20	2/23
Revised sensitivity warning	1	0	12	10/23
Revised combined warning	1	0	12	10/23

Separating conclusion-changing scenarios from standalone fit diagnostics reduces warnings without a reference change from 20 to 12 while retaining the single detected reference event. Precision increases from 4.8% to 7.7%, and cases receiving no warning increase from 2/23 to 10/23. MAE among cases receiving no warning is 13.86 points, compared with 23.28 for the original rule and 16.36 across all conditions. That subset includes any false negatives and is not a reliability certificate. The comparison is retrospective, with one reference event, and the remaining warnings quantify the inspection burden for laboratory calibration. All original readings remain in the fits.

Paired computational A/B comparison

Arm A uses the isotonic conclusion with no warning. Arm B uses the same conclusion with the revised combined warning and evidence report. The legacy rule provides an ablation of standalone fit diagnostics. All arms share predictions, cases and reference events. The comparison measures missed changes, warnings without a reference change, the fraction of cases receiving no warning and their prediction error. It cannot measure whether scientists make better decisions. The demo switches between A and B report views without changing the underlying fit. All 311 case-level decisions across the real and three simulated partitions are exported in `paired_ab_decisions.csv`.

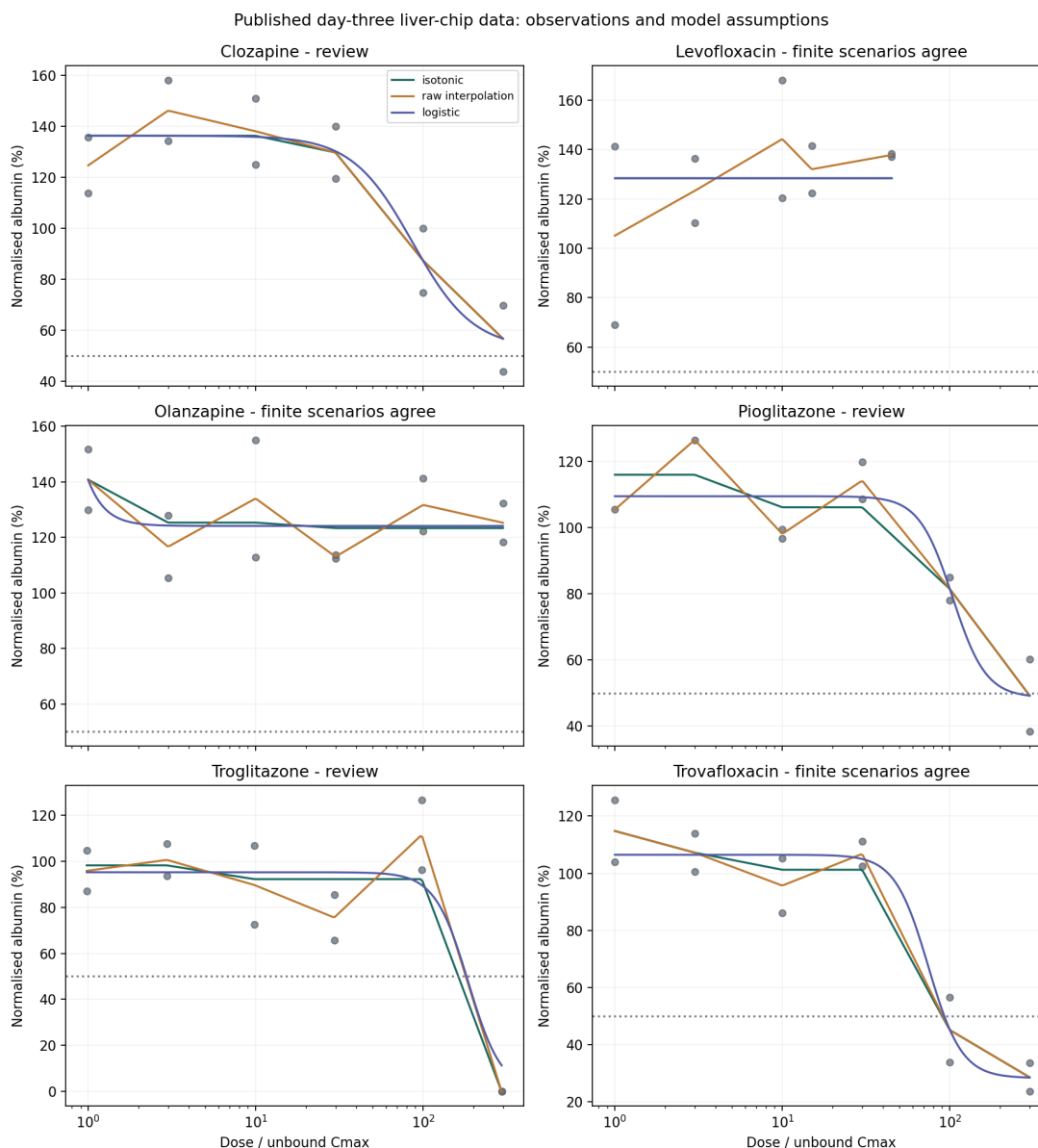
Real-data case studies

The full-data isotonic crossings are approximately 162.38 times unbound C_{max} for troglitazone, 89.99 for trovafloxacin and 292.51 for pioglitazone. Clozapine, olanzapine and levofloxacin are right-censored in their primary fits: the operational crossing is not observed in the tested range.

Clozapine/olanzapine. Clozapine's conclusion changes after deleting one influential reading. Olanzapine has no material crossing change in the tested scenarios, but its logistic fit has a parameter diagnostic. The interface distinguishes those two reasons rather than treating every warning as evidence of a changed biological response.

Troglitazone/pioglitazone. Both primary fits have finite crossings. Troglitazone is sensitive to dose deletion. Pioglitazone has two missing albumin readings and changes under reading/dose deletion; the displayed missing-value probes do not produce a material change under the fixed rule. That absence of change does not resolve its missingness, because the probes are finite hypothetical scenarios rather than exhaustive bounds.

Trovafloxacin/levofloxacin. Trovafloxacin's primary crossing lies inside the measured range; levofloxacin is right-censored. Neither has a material crossing change in the tested scenarios, but both have logistic fit diagnostics. These endpoint contrasts do not independently validate a clinical safety ranking.



Observed readings and fitted alternatives for all six compounds

Synthetic and engineering evaluation

The simulator uses the original development and evaluation seeds, 20261001 and 20261002, and a fresh revision-evaluation seed, 20261004, with 96 experiments in each partition. The revision was frozen before the fresh seed was generated; no rule was changed after inspecting its results. The original two partitions are retrospective for this revision. The generator alternates Hill and piecewise latent responses across eight scenarios: clean measurements, random missingness, response-dependent missingness, truncated dose range, singleton counts, an influential reading, non-monotonic response and a run shift. Each simulated experiment is analysed independently; latent responses are reserved for scoring.

The fresh seed contains ten experiments with an incorrect crossing state or a location error greater than a factor of two under the defined latent reference. Its paired results are:

Fresh simulations, 96 identical cases	Detected errors	Missed errors	False warnings	Cases with no warning
A, curve only	0	10	0	96/96
Original diagnostic-inclusive rule	10	0	67	19/96
B, revised guarded report	10	0	52	34/96

Demoting standalone diagnostics reduces false warnings by 15 while retaining detection of all ten specified errors in this seed, giving precision of 16.1% and recall of 100%. The 52 remaining false warnings identify the workload that a laboratory adaptation must measure and calibrate. These experiments test defined generator scenarios; independent biological validation requires laboratory cases.

On the earlier evaluation seed, the revised rule detects eight of ten errors, misses two and produces 52 false warnings, compared with 65 for the original rule. Detection therefore varies across the tested seeds. Both outcomes are retained in `legacy_results_20261002.json` and `paired_ab_comparison.csv` so an adopter can inspect the full comparison.

Separate engineering fixtures cover clean input, incompatible dose units, incompatible response units, a duplicated source record, a derived summary and an unsafe pairing request. All six fixture expectations pass. The checks also verify every extracted source cell, row/ID invariance of the endpoint result, crossing boundaries and held-dose separation. Six constructed fixtures are not a population-scale estimate of fault-detection accuracy.

Local language model and action safeguards

The local planner separates action selection from numerical evidence. The optional SmoLM2-135M-Instruct model, with 134,515,008 parameters, selects one of eight action classes. Inputs are the calculated conclusion state, dose range, changed-scenario counts, concern counts, researcher request, resource limits and eligible codes. Full source cards bypass model inference. The runtime scores each short code by mean token log likelihood and selects the highest score. Eligibility checks examine blocked input, recognised unsupported requests and resources; forbidden choices fall back to request-aware rules. Python copies the first relevant existing card, including its numerical evidence and source/scenario references. The model selects the action class; the renderer supplies the named condition and explanation from the audit. [SmoLM2 model card](#), Allal et al., 2025.

The policy and engineering gate were specified before model inference. Training contains 728 authored requests from 96 simulated experiments; validation contains 182 from 24 experiments. All requests and duplicated-input variants from one experiment remain in its partition. The 364 test requests come from 48 held-out experiments, including new piecewise response shapes and non-monotonic/run-shift cases. Six published compounds provide 42 further demonstration requests, excluded from adaptation and checkpoint selection. Targets come from an explicit heuristic policy, not independent scientist adjudication or biological truth. Request templates and policy were authored together.

Full supervised adaptation uses assistant-code and EOS loss only, three epochs, batch size 8, learning rate 0.00005, AdamW weight decay 0.01 and gradient clipping at 1.0. The pinned base revision is `12fd25f77366fa6b3b4b768ec3050bf629380bac`. Training took 334.34 seconds on the local RTX 3050 6 GB laptop GPU with BF16 autocast. Epoch 2 was selected by lowest validation assistant-token loss; no test outcome selected the checkpoint. The saved deployment weights are FP32.

Planner	Raw agreement with test policy	Delivered agreement after guard
Fixed action order	237/364 (65.1%)	Not evaluated
Request-aware rules	364/364 (100%)	No correction needed
Untuned 135M model	72/364 (19.8%)	144/364 (39.6%)
Adapted 135M model	337/364 (92.6%)	361/364 (99.2%)

The scope guard corrects all 24 tested unsupported-request errors, which proposed INSPECT. All 76 protected test cases receive the correct delivered action, and every rendered source/scenario reference is valid. Three delivered discrepancies remain: two model-review requests select INSPECT rather than COMPARE, and one finite-crossing request selects RECOVER rather than REPORT. No raw action is resource-ineligible; eligibility therefore needs a separate check of action priority. Protection for arbitrary new request wording requires further evaluation.

The predefined gate requires at least 90% raw agreement overall and 80% in every represented action class. Overall agreement reaches 92.6%, while raw ABSTAIN agreement is 24/48 (50%) and COMPARE is 0/2, leaving the per-class gate unmet. These classes are priorities for the next training and evaluation set.

Other raw action counts are FIX 28/28, INSPECT 66/66, RECOVER 21/21, REPEAT 44/44, EXTEND 4/4 and REPORT 150/151. Rare-class counts are too small to establish reliable class performance. The 364 requests are grouped within 48 experiments and must not be treated as 364 independent experimental observations. Published demonstrations match 41/42 policy targets; pioglitazone's finite-crossing request selects RECOVER where the policy prioritises REPEAT. Rules match all 42. This is a policy discrepancy rather than evidence of biological harm.

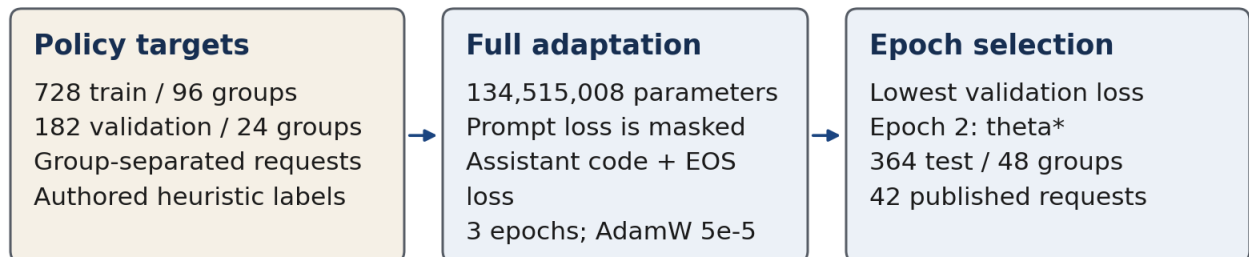
Adapted CPU model scoring on an Intel Core Ultra 5 125H with 31.7 GiB RAM and four Torch threads takes a warm median of 1.33 seconds and p95 of 2.29 seconds. Scoring includes prompt prefill and code likelihoods, excluding tokenisation, numerical auditing and rendering. Cached model loading takes 1.17 seconds, excluding package imports. Process peak resident memory is 1.58 GiB; weight files occupy 513.2 MiB. PyTorch 2.8.0+cu126 and Transformers 4.57.1 were used for CPU deployment. Quantisation and deployment on a named edge device are not implemented or measured.

The prototype demonstrates how a compact language model can select an action while numerical calculations, source references and eligibility checks remain in code. Request-aware rules match all authored test labels and provide the default comparison; the reported model score measures policy agreement, not superiority over rules. A laboratory adapting the planner can replace the authored policy targets with scientist-reviewed requests and actions, retrain or compare models, and retain the same evidence checks. New held-out experiments are needed after tuning. Protocol, training history and every decision are preserved under `small_model/`.

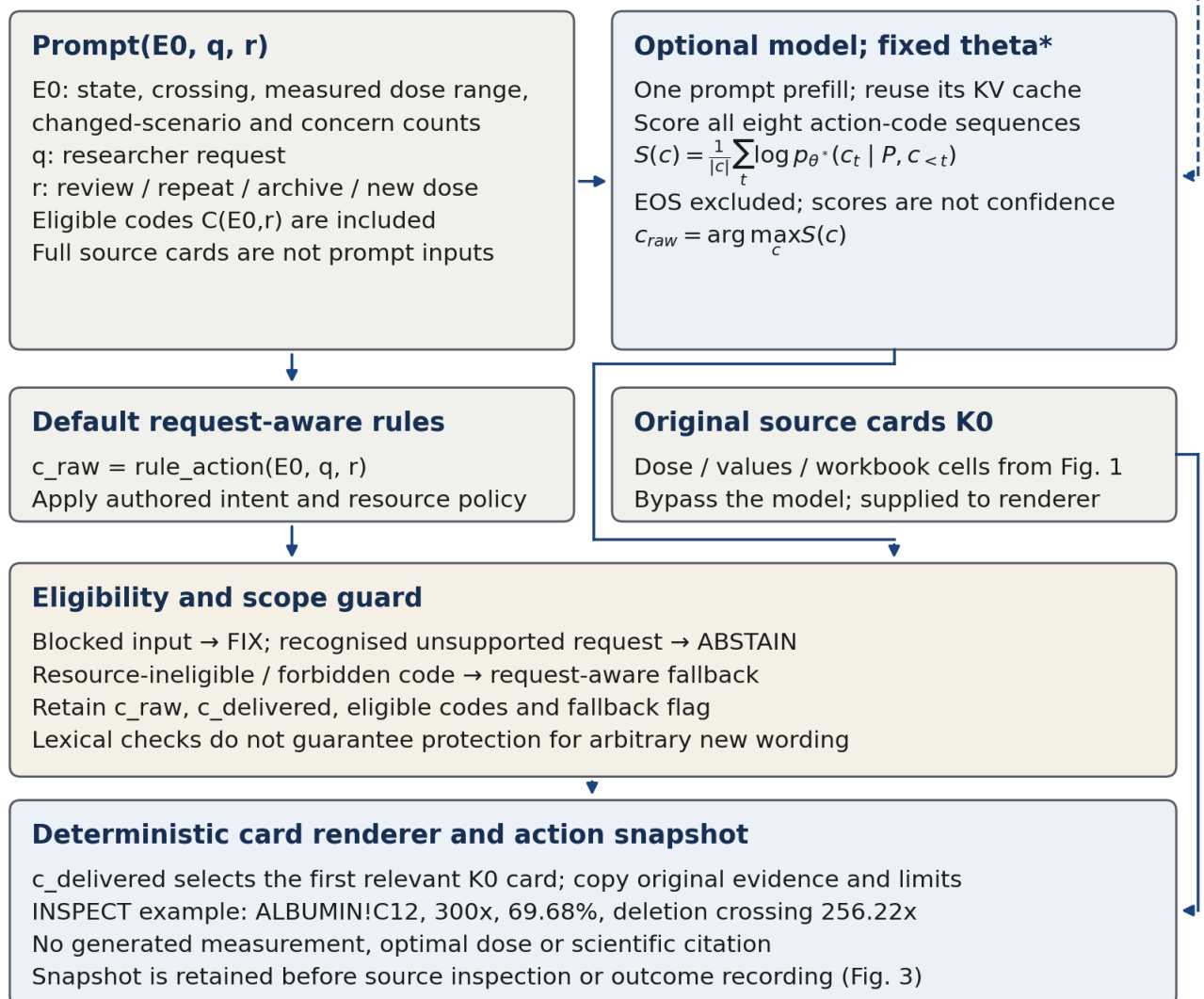
Source-specific action selection

Figure 2. Adaptation is separate from inference; scientific values bypass the model

A. Offline supervised adaptation of SmolLM2-135M



B. Runtime inputs and engine choice



Codes: FIX / INSPECT / REPEAT / RECOVER / COMPARE / EXTEND / REPORT / ABSTAIN

Figure 2. Group-separated supervised adaptation, fixed-parameter runtime action-code scoring, eligibility checks and deterministic rendering. The dashed arrow assigns the selected checkpoint to inference. Scientific values and source cards bypass the language model.

At runtime, one prompt prefix provides the key/value cache used to score all eight action codes. Let u be the prompt containing the audit summary, request and resources; θ denotes the fixed adapted model parameters. For an action code c , let T_c be its number of tokenizer tokens, c_t its token at position t , and $c_{<t}$ its preceding tokens. The score and raw selection are

$$S(c) = \frac{1}{T_c} \sum_{t=1}^{T_c} \log p_{\theta}(c_t | u, c_{<t})$$

$$\hat{c}_{\text{raw}} = \arg \max_{c \in C} S(c).$$

The set C contains FIX, INSPECT, REPEAT, RECOVER, COMPARE, EXTEND, REPORT and ABSTAIN. Raw selection ranks all eight codes before scope and resource checks determine delivery. Runtime scoring excludes the end-of-sequence token (EOS); supervised adaptation includes the assistant code and EOS while masking prompt tokens. This length-normalised likelihood ranks action codes and is not confidence in the scientific result. `small_model/model_runtime.py` implements scoring; `small_model/planner.py` constructs the summary, applies the checks and renders the card. Default rules use the same checks and renderer.

From a recommendation to a recorded outcome

A scientist reviews the endpoint result and affected source readings, then specifies a goal and available resources. The planner delivers an eligible action and a source-specific explanation. "Record what happened" opens the outcome form. The action is saved before source inspection; returning restores the plan and unfinished draft.

P0 retains the original records and audit, request/resources, raw and delivered action codes, inference details and outcome history. A canonical signature of sorted record contents matches the evidence version, rather than relying on the compound name alone. The workflow identifier is a software record identifier; it is not an invented chip ID. Plans and drafts are kept in the current browser tab's session storage, and the complete JSON export preserves them outside that session.

There are two distinct outcome paths. A note with a source reference documents inspection, unresolved issues or reporting. It leaves R0, A0 and every computed warning unchanged. Alternatively, a complete compatible updated CSV R1 can be submitted with a reference and a researcher statement about compatibility. The server checks same-compound day-three normalised albumin, rejects duplicate/derived evidence, retains every baseline source-record ID and rejects conversion of an observed original to missing. It reports changed record fields without overwriting the baseline. The supplied reference and laboratory compatibility are not independently verified.

Delivered action	Required evidence for a measurement-related outcome
RECOVER	An originally missing record becomes observed under its existing source ID.
REPEAT	An additional original observed record has a new source ID at the condition named in the action. The software cannot prove an independent chip.
EXTEND	An additional original observed record is supplied at a dose absent from the baseline's observed doses. The software does not choose an optimal dose.
FIX	The baseline is blocked and the documented revised input passes the source gate.
INSPECT	Record a finding/reference; a documented compatible evidence revision can be reassessed.
COMPARE / REPORT / ABSTAIN	Document interpretation, reporting or the reason for declining the request. Notes do not constitute new measurements.

Metadata changes alone do not satisfy recovery, repeat or dose-extension requirements. Rejected uploads preserve the prior record and explain the rejection. Accepted revisions recompute $A0 = F(R0)$ and $A1 = F(R1)$ using the unchanged auditor and compare record/check changes, crossing state and location, and the measured range. Both evidence versions remain available. Open updated Review, export the history, or start another action from A1. Each action retains one reassessment; further revisions start a new plan.

Outcome recording and evidence reassessment

Figure 3. Notes document a decision; a compatible CSV starts a separate evidence version

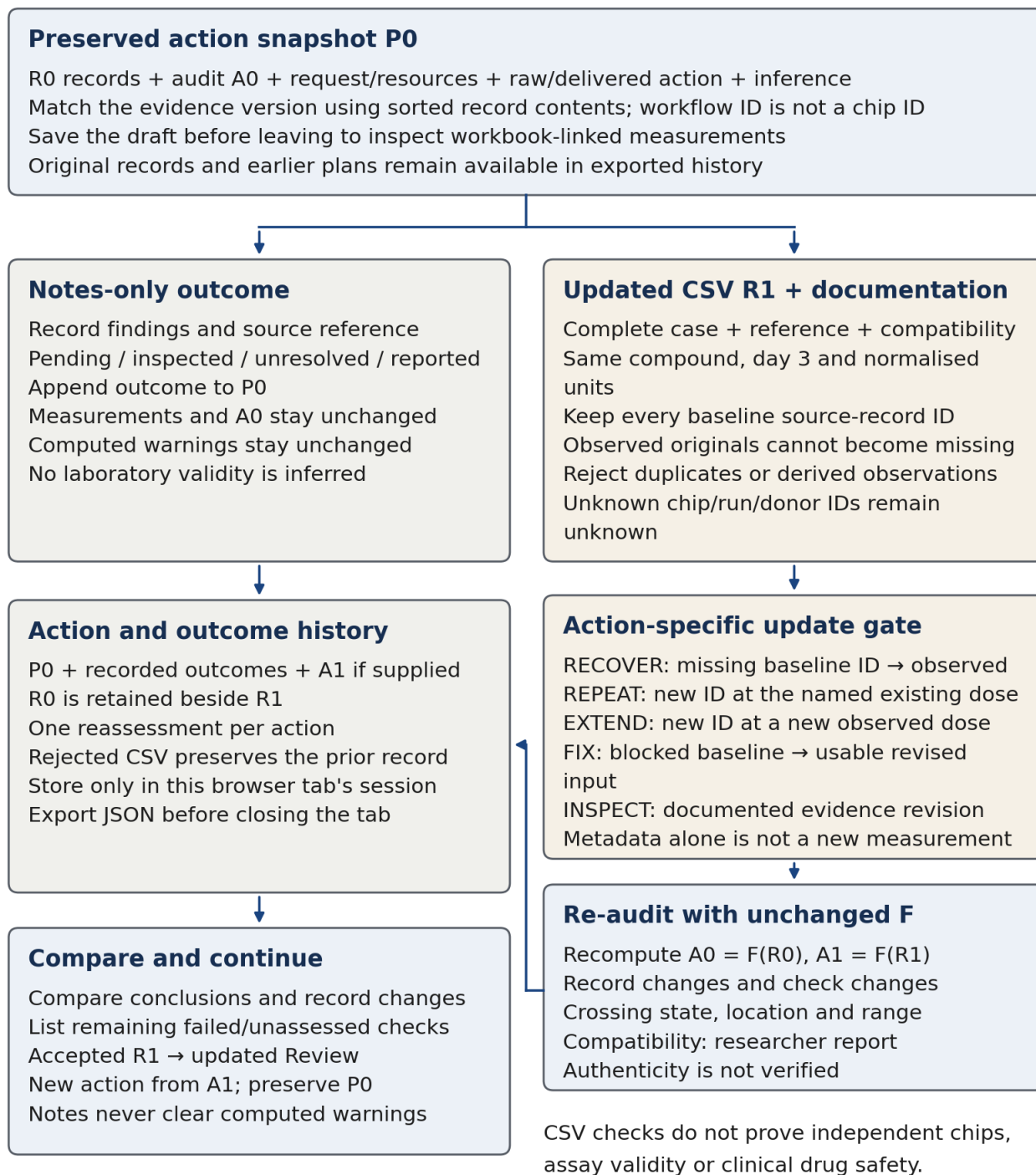


Figure 3. Preserved action, notes-only outcome and action-specific CSV reassessment. Accepted revised evidence is compared with the retained baseline using the same auditor. Recording a note never clears a computed warning.

Concrete source-linked example and engineering checks

For published Clozapine, the original ALBUMIN!C12 value is 69.67854% at 300 times unbound Cmax. The full primary fit is right-censored; temporarily deleting that reading produces a crossing at approximately 256.22 times unbound Cmax. The source-inspection action names this cell, value and dose. Recording that the value was inspected does not remove the reading or resolve its numerical influence.

An explicitly synthetic test fixture changes C12 to 40% to exercise the documented-revision path. Reassessment changes the crossing from right-censored to approximately 246.54 times unbound Cmax and retains the original alongside the changed record. This value is a software test, not a new laboratory measurement or a correction to Ewart et al. Separate synthetic fixtures exercise missing-value recovery, same-dose repeats and new-dose additions, and reject false completion, removed originals, observed-to-missing substitutions and invalid units.

Browser checks exercise actual local-model inference, leaving to inspect a source and returning, draft restoration by evidence version, notes-only recording, rejected updates, accepted synthetic revisions, before/after navigation, subsequent planning, JSON export and unavailable session storage. These checks verify the demonstrated path from a recommendation to a recorded outcome. Laboratory and researcher evaluation are separate adoption checks.

Demonstration and reproduction

Set up the environment, reproduce the analysis and start the app:

```
conda env create -f environment.yml
conda activate ai4s-liver-auditor
python run_auditor.py
python check_auditor.py
python check_followthrough.py
python -m small_model.check_planner
python demo.py
```

Open the local URL printed by `demo.py`. To use the saved viewer without Python, open `auditor_demo.html`. Public source files are downloaded if absent. Evaluation outputs are written under `results/auditor/`.

For the model planner, extract the separately supplied `liver_chip_slm_result.zip` into the code package's project root and create the environment from `small_model/environment.yml`. Run `python demo.py --model small_model/checkpoints/smollm2-135m-policy --threads 4`, open `planner.html` on the printed local server and select the local model. A Colab L4 notebook and upload ZIP provide an optional reproduction route; that notebook has not been executed on Colab.

The saved run in `results/auditor/results.json` records approximately 0.83 seconds for the real-data analysis and 12.54 seconds for the full analysis including all three simulation partitions, excluding plotting/export. Python 3.12.14, NumPy 2.3.5 and SciPy 1.16.3 were used on this Windows CPU.

A fresh package extraction was checked in an isolated Python 3.12 environment with pinned core dependencies. It regenerated the real and three simulated partitions. Source-cell, input, fold, action-policy and synthetic outcome checks passed. Counts, warning decisions, source records and primary fits matched saved results; floating metrics were compared at relative and absolute tolerances of 0.000001. Dependency versions and check results are in `submission/reproduction_checks.json`.

The extracted checkpoint was checked in the existing pinned model environment: cached code scores matched complete forwards, and assistant-only training labels passed checks. This check did not reinstall the model environment or repeat training. Browser outcome checks used the adapted model and synthetic evidence revisions.

Laboratory adaptation and validation

The released code and source files let a laboratory reproduce the reference experiment and inspect every calculation. The current CSV interface accepts day-three albumin on the documented percentage and unbound-Cmax scales. For another assay, laboratories make coordinated code changes to the following components and validate the adaptation.

Component	Implementation to adapt	Laboratory-specific decision
Source records	<code>prepare_data.py</code> , CSV handlers in <code>demo.py</code> and <code>followthrough.py</code>	Map original observations, missingness, units and source references; preserve documented chip, donor and run identities.
Endpoint and crossing	<code>auditor.py</code> and <code>model.py</code>	Define the response scale, timepoint, dose scale, curve assumptions and meaningful threshold. The reference case uses a decreasing response and 50% crossing.
Evidence review	<code>audit_sources</code> , <code>evidence_checklist</code> and <code>analyse_evidence</code> in <code>auditor.py</code>	Specify required controls, input checks, sensitivity scenarios and what change warrants inspection.
Follow-up policy	<code>small_model/planner.py</code> and model training targets	Define permitted actions, resource constraints and scientist-reviewed priorities; update the evidence summary and renderer together.
Outcome recording	<code>followthrough.py</code> and the outcome interface	Define the evidence needed for each action, retain the baseline and compare documented revisions.

Validate source records against the laboratory's original export, document chip/run identities and controls, and hold out experimental groups relevant to the intended use. Compare endpoint predictions and conclusion changes on untouched groups, measuring detected errors, missed errors and inspection burden. The reference evaluation here measures evidence restoration and defined simulation scenarios; clinical assessment requires an independent reference.

For planner adaptation, use scientist-reviewed requests and action labels, group partitions by experiment, and report raw and delivered decisions by class. Include resource restrictions and requests outside the assay's scope. Retain the numerical renderer and eligibility checks when comparing models. Laboratory and reader studies, with independent adjudication and controlled case order, remain adoption evaluations beyond this submission.

Observed false positives and missed events provide warning-calibration targets. The earlier pilot's nominal percentile resampling ranges covered 8/23 held-dose means; the app therefore shows explicit sensitivity scenarios without claiming calibrated 95% intervals.

The submission delivers an executable workflow from a dose-response conclusion to a source-specific follow-up and a recorded outcome. Published liver-chip data provide the reproducible reference case. Statistical calculations, a compact action-selection model and code-level checks have separate roles, allowing laboratories to adapt the assay and policy while retaining evidence provenance and researcher review.

External tools and attribution

Data attribution follows Ewart et al. and CC BY 4.0. NumPy, SciPy, openpyxl and Matplotlib provide numerical, workbook and plotting functions. PyTorch and Transformers support adaptation of HuggingFaceTB/SmolLM2-135M-Instruct under Apache 2.0; its model card and licence accompany the checkpoint. Codex assisted implementation and writing. Undermind and Parallel Search assisted literature discovery and source verification.

Appendix A. Fitted parameters and diagnostics

Compound	Logistic lower / amplitude / log-midpoint / slope	Diagnostics
Clozapine	52.71 / 83.69 / 1.937 / 5.527	None
Levofloxacin	128.5 / 1.752e-10 / 3.642 / 0.1	parameter_at_bound, weak_parameter_identification
Olanzapine	124.1 / 522.3 / -0.3417 / 10	parameter_at_bound, weak_parameter_identification
Pioglitazone	48.64 / 60.76 / 2.017 / 10	parameter_at_bound
Troglitazone	9.322e-07 / 95.17 / 2.27 / 10	parameter_at_bound
Trovafloxacin	28.22 / 78.24 / 1.872 / 10	parameter_at_bound

Active parameter bounds remain visible diagnostics. They trigger review alone in the legacy comparator, but not in the revised rule. Failed model comparisons and material changes still trigger review.

Appendix B. Material scenario changes

Compound	Scenario	Evidence removed or model changed	Result / crossing
Clozapine	reading_deletion	dose=300;reading=1	within_range / 256.22
Pioglitazone	reading_deletion	dose=300;reading=2	right_censored / censored
Pioglitazone	dose_deletion	dose=300	right_censored / censored
Troglitazone	dose_deletion	dose=98.2	within_range / 64.14
Troglitazone	dose_deletion	dose=294.72	right_censored / censored

All scenarios, including unchanged probes, are exported in results/auditor/conclusion_scenarios.csv.

Source-linked influential observations

Compound	Original source cell / value	All readings	Temporary deletion
Clozapine	ALBUMIN!C12 / 69.68%	right_censored	within_range / 256.22
Pioglitazone	ALBUMIN!C49 / 38.37%	within_range	right_censored / censored

Original observations remain in the fit. Unavailable controls, calibration and device evidence are marked Not assessed.

Appendix C. All 23 held-dose predictions

Compound	Dose	Observed mean	Isotonic	Raw	Logistic	Restoration change	Review
Clozapine	3	146.26	131.44	131.13	131.33	False	True
Clozapine	10	138.14	132.52	137.65	135.06	False	True
Clozapine	30	129.80	113.01	113.93	135.71	False	True
Clozapine	100	87.40	91.58	91.58	61.15	False	True
Levofloxacin	3	123.26	129.80	123.79	129.80	False	False
Levofloxacin	10	144.25	124.56	129.81	124.56	False	False
Levofloxacin	15	132.02	127.61	142.52	127.61	False	False
Olanzapine	3	116.62	137.55	137.55	140.87	False	False
Olanzapine	10	134.02	121.65	114.74	121.63	False	False
Olanzapine	30	113.03	127.45	132.92	126.90	False	False
Olanzapine	100	131.70	119.15	119.43	122.21	False	False
Pioglitazone	3	126.39	105.92	101.86	105.97	False	True
Pioglitazone	10	98.01	114.95	119.97	115.80	False	True
Pioglitazone	30	114.11	90.11	90.11	98.33	False	True
Pioglitazone	100	81.45	76.36	80.20	108.22	False	True
Troglitazone	2.95	100.53	94.06	92.84	93.75	False	True
Troglitazone	9.82	89.64	95.70	87.45	96.63	False	True
Troglitazone	29.5	75.50	99.34	100.03	100.34	False	True
Troglitazone	98.2	111.41	36.03	36.03	18.96	True	True
Trovafoxacin	3	107.20	108.31	105.66	106.20	False	False
Trovafoxacin	10	95.65	106.97	106.97	110.90	False	False
Trovafoxacin	30	106.77	71.52	71.52	72.66	False	False
Trovafoxacin	100	45.08	63.20	65.85	103.71	False	True

Responses are albumin percentage points; doses are multiples of unbound Cmax. The constant baseline and training-dose lists are in the exported CSV. All readings at the held dose were excluded.

Appendix D. Synthetic evaluation by scenario

Scenario	Experiments	Reference errors	Detected	Missed	False warnings
clean	12	0	0	0	4
influential_reading	12	5	5	0	5
non_monotonic	12	1	1	0	11
random_missing	12	1	1	0	11
response_dependent_missing	12	3	3	0	9
run_shift	12	0	0	0	2
singletons	12	0	0	0	7
truncated_range	12	0	0	0	3

Fresh revision seed 20261004; revised combined warning. Each scenario contains both Hill and piecewise shapes. Latent truth includes the generated run shift. Truth is evaluated within the observed dose range, so a truncated curve can correctly be censored.

Paired A/B comparison on identical cases

Cases	Arm	Detected / missed	False warnings	Cases with no warning
Real, retrospective	A: curve only	0 / 1	0	23 / 23
Real, retrospective	Legacy: diagnostics trigger review	1 / 0	20	2 / 23
Real, retrospective	B: guarded report	1 / 0	12	10 / 23
Fresh simulations	A: curve only	0 / 10	0	96 / 96
Fresh simulations	Legacy: diagnostics trigger review	10 / 0	67	19 / 96
Fresh simulations	B: guarded report	10 / 0	52	34 / 96

All arms share fitted predictions. This computational comparison does not establish an effect on human decisions. Earlier inspected partitions and case-level decisions are retained in the bundle.

Appendix E. Engineering fixtures

Fixture	Expected blocked code	Detected codes	Pass
clean	None	None	True
wrong_dose_unit	incompatible_dose_unit	incompatible_dose_unit	True
wrong_response_unit	incompatible_response_unit	incompatible_response_unit	True
duplicate	duplicate_source_record	duplicate_source_record	True
derived	derived_observation	derived_observation	True
unsafe_pairing	unsupported_pairing	unsupported_pairing	True

Appendix F. Real albumin source records

All 70 source readings follow. Workbook: Supplementary Data 8, ALBUMIN worksheet, DOI 10.1038/s43856-022-00209-1. Experimental IDs remain unknown. Source values and cells were checked individually.

Compound	Source cell	Dose x unbound Cmax	Albumin %	Missing
Clozapine	C2	1	135.709948005764	False
Clozapine	C3	1	113.769799561393	False
Clozapine	C4	3	134.233207172339	False
Clozapine	C5	3	158.282985283381	False
Clozapine	C6	10	125.161799672409	False
Clozapine	C7	10	151.110244505707	False
Clozapine	C8	30	140.140170283521	False
Clozapine	C9	30	119.465799728161	False
Clozapine	C10	100	100.057206839291	False
Clozapine	C11	100	74.7416510617572	False
Clozapine	C12	300	69.6785397282349	False
Clozapine	C13	300	43.730095117456	False
Olanzapine	C14	1	151.743133338953	False
Olanzapine	C15	1	129.802984894581	False
Olanzapine	C16	3	105.331280894708	False
Olanzapine	C17	3	127.904318172325	False
Olanzapine	C18	10	155.118540894634	False
Olanzapine	C19	10	112.925947783732	False
Olanzapine	C20	30	113.769799561393	False
Olanzapine	C21	30	112.293058727967	False
Olanzapine	C22	100	122.208318228077	False
Olanzapine	C23	100	141.194985005597	False
Olanzapine	C24	300	132.334540450083	False
Olanzapine	C25	300	118.20002183915	False
Troglitazone	C26	0.98	86.9413297363926	False
Troglitazone	C27	0.98	104.619181352019	False
Troglitazone	C28	2.95	93.4542223314824	False

Compound	Source cell	Dose x unbound Cmax	Albumin %	Missing
Troglitazone	C29	2.95	107.596503757495	False
Troglitazone	C30	9.82	106.666090561414	False
Troglitazone	C31	9.82	72.6129657156675	False
Troglitazone	C32	29.47	65.5418248914015	False
Troglitazone	C33	29.47	85.4526685336544	False
Troglitazone	C34	98.2	96.2454621422464	False
Troglitazone	C35	98.2	126.576933981147	False
Troglitazone	C36	294.72	0.0	False
Troglitazone	C37	294.72	0.0	False
Pioglitazone	C38	1	105.363511953388	False
Pioglitazone	C39	1	Missing	True
Pioglitazone	C40	3	126.390851386435	False
Pioglitazone	C41	3	Missing	True
Pioglitazone	C42	10	99.4088671424351	False
Pioglitazone	C43	10	96.6176273316712	False
Pioglitazone	C44	30	119.691875974113	False
Pioglitazone	C45	30	108.526916953577	False
Pioglitazone	C46	100	84.8944205269977	False
Pioglitazone	C47	100	78.0093625199635	False
Pioglitazone	C48	300	60.1454280871055	False
Pioglitazone	C49	300	38.3737580526897	False
Trovafloracin	C50	1	125.68987893106	False
Trovafloracin	C51	1	103.881661404813	False
Trovafloracin	C52	3	113.930545863958	False
Trovafloracin	C53	3	100.46076463394	False
Trovafloracin	C54	10	86.1357590999432	False
Trovafloracin	C55	10	105.16449786078	False
Trovafloracin	C56	30	111.151067321069	False
Trovafloracin	C57	30	102.385019095371	False
Trovafloracin	C58	100	56.4167176519772	False
Trovafloracin	C59	100	33.7532758217525	False

Compound	Source cell	Dose x unbound Cmax	Albumin %	Missing
Trovafloracin	C60	300	23.4905852866127	False
Trovafloracin	C61	300	33.5394699682774	False
Levofloxacin	C62	1	141.194985005597	False
Levofloxacin	C63	1	69.0456508949894	False
Levofloxacin	C64	3	136.34283683901	False
Levofloxacin	C65	3	110.183429061296	False
Levofloxacin	C66	10	168.198244560971	False
Levofloxacin	C67	10	120.309651505821	False
Levofloxacin	C68	15	141.616910894428	False
Levofloxacin	C69	15	122.419281172492	False
Levofloxacin	C70	45	137.18668883919	False
Levofloxacin	C71	45	138.452466505681	False

Appendix G. Frozen protocol and software

Frozen evaluation protocol

Frozen on 2 October 2026, before implementing the auditor comparisons. The Ewart workbook has already been inspected; real-data evaluation is exploratory, not blinded.

The primary endpoint is source-normalised day-three albumin. The operational threshold is 50%. Doses are multiples of unbound C_{max}. Six compounds form three published comparison pairs. Experimental chip/donor/run identities are unknown.

Models and conclusion

Compare weighted non-increasing isotonic regression, raw dose-mean interpolation, a constant training mean, and a four-parameter decreasing logistic curve. Interpolation is linear in log₁₀ dose. Fits use dose means weighted by observed counts. Logistic lower level and amplitude are non-negative, slope is between 0.1 and 10, and log midpoint lies within two decades of the training dose range. Fit failure, active parameter bounds and rank deficiency are reported. A fit's midpoint is not automatically its 50% normalised-response crossing.

A crossing conclusion comprises left censoring, right censoring or a crossing within the tested range. A response at or below 50% at the lowest tested dose is left-censored under the implementation's boundary convention. Otherwise, report the first downward crossing; right censoring means no such crossing was reached within the tested range and does not assert an eventual crossing. A material location change is a change greater than a factor of two. This is a fixed operational review rule, not a clinical margin. Report continuous changes as well. Raw interpolation can be non-monotonic; flag multiple downward crossings or transitions across 50% in both directions.

Implementation correction, 5 October 2026: search interior downward crossings before assigning right censoring to a raw curve whose final response exceeds 50%. Add a diagnostic for a downward crossing followed by a return above 50%. Thresholds and sensitivity scenarios remain unchanged.

Auditor scenarios

The primary fit is isotonic. Refit it after deleting each observed reading and after deleting each dose, including boundaries. Compare it with raw interpolation and a fitted logistic curve. Do not include the constant baseline in the model-agreement warning. Hypothetical missing albumin values are 0, 50, 100, 150 and 200 percent, one missing record at a time. These values are sensitivity probes, not imputations or biological bounds. The missingness remains unresolved beyond this finite scenario set.

Report model disagreement, observation/dose deletion sensitivity and missing-value sensitivity separately. The combined review decision is their union, plus failed/weakly identified logistic comparison. Source faults are a separate output. Missing experimental identity limits pairing and group validation but does not force every endpoint conclusion into review.

Real-data evaluation

Use all 23 interior dose conditions. Remove all readings at a held dose before prediction and warning computation. Score the held observed mean in percentage points using MAE and RMSE; report every model's eligible count and failures. Do not extrapolate in this prediction test.

After fitting and warning, restore the withheld dose and compare the isotonic conclusion to its full-data counterpart. A changed state or greater-than-twofold crossing-location change is the reference event. The reference is evidence restoration, not biological ground truth. Report precision, recall, false warnings, missed changes, retained fraction and held-dose error among retained conditions. Undefined ratios remain null.

Compare no warning, training missingness or fewer-than-two-readings warning, source-only warning, sensitivity-only warning and their combination. Report per-compound results. The 23 folds are correlated within six compounds; do not treat them as 23 independent drug studies or make significance claims.

Synthetic evaluation

Use independent seeds 20261001 (development) and 20261002 (evaluation), 96 experiments per partition. Generate six log-spaced doses, two readings per dose, and explicit distinct synthetic chips within each simulated run. Cycle through Hill and piecewise response shapes and eight scenarios: clean, random missingness, response-dependent missingness, truncated dose range, singleton counts, influential observation, non-monotonic response and run shift. There is no longitudinal or multi-endpoint pairing claim. The latent response including the simulated run shift determines the reference crossing.

Analyse each synthetic experiment independently. Do not train on the evaluation experiments or change thresholds using their results. Compare full, source-only, sensitivity-only and count-warning outputs against the known crossing state/material location error. Separate controlled metadata-fault fixtures from response simulation, include a clean fixture, and report only those known fault cases. Simulation cannot establish human injury prediction.

Reproduction and changes

Use a dedicated CPU Conda environment and pinned dependencies. Preserve the original pilot output. Save scenario rows, source issues, model predictions, confusion counts, undefined quantities, generator assumptions and measured runtime. Any deviation from this protocol must be documented before presenting its result; negative results are retained.

Revision frozen on 4 October 2026

The 2 October real-data and synthetic results have already been inspected. Preserve their saved output as `results/auditor/legacy_results_20261002.json`. Comparisons on those cases are retrospective. The revision changes evidence reporting and warning composition; fitted curves, the 50% threshold, the factor-of-two rule, scenarios and generator remain unchanged.

Separate hard input failures, missing evidence, material sensitivity changes and numerical fit diagnostics. The revised sensitivity warning is the union of material scenario changes. The revised combined warning adds missing-source warnings. Logistic parameter bounds and weak identification remain visible diagnostics but no longer trigger review alone. A failed logistic comparison still counts as a material model scenario. Preserve the previous diagnostic-inclusive rule as a legacy comparator. Do not change these rules after observing the new results.

For a paired computational A/B comparison, A uses the same fitted isotonic conclusion without warnings; B uses the revised combined warning and evidence report. Include the legacy rule to expose the effect of demoting standalone diagnostics. Compare missed reference changes, warnings without a reference change, retained fraction and held-dose error among retained cases. The A/B arms share predictions, so an improvement in raw prediction accuracy is not expected. This is not a human reader study or a biological treatment-control experiment.

After implementing the frozen revision, generate 96 additional experiments with seed 20261004, partition `revision_evaluation`. Apply A, legacy and revised rules to identical cases before accessing latent truth for scoring. This seed has not been inspected when the revision is specified. It is fresh simulated evaluation within the existing generator, not independent experimental data or a new response family. Report all results, including losses in recall and false warnings. Retain the earlier two partitions with explicit retrospective labels.

Checklist statuses are Pass, Fail and Not assessed. A pass concerns the named computational check only. Missing raw vehicle/reference controls, assay calibration, device QC or documented experimental independence

remain Not assessed. A failed threshold-bracketing check prevents a finite in-range estimate; it does not invalidate a correctly censored result. Do not combine checklist statuses into a safety score.

Attach source-record IDs, source cells and removed values to observation-deletion scenarios. Keep original readings in fitting/export; temporary deletion measures influence and does not establish an erroneous outlier. Highlight observations whose deletion changes the operational conclusion. Exclusion requires separately documented technical evidence; this release does not automatically exclude observations. Export all paired decisions and checklist items, not just favourable cases.

Implementation clarification: reading-deletion checks require another observed reading at that dose. Dose-deletion checks require at least three observed doses before deletion. Singleton-dose influence is covered by those eligible dose-deletion scenarios. These eligibility conditions are unchanged from the original implementation.

Artifact generation uses ReportLab 4.4.9, separate from the numerical tool. Reviewers need only the core dependencies to rerun the experiment or demo.